Abdul Moiz

Ph.D. Civil Engineering

Hydrological Modeler | Water Resources Engineer

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Experience

Research & Development Engineer III (Hydrology & Water Resources) — **January 2, 2025 – Present**
University of California San Diego — San Diego, CA, U.S.
Center for Western Weather and Water Extremes (CW3E) | Scripps Institution of Oceanography

- Leading research integrating snow hydrometeorological datasets and long-range ensemble forecasts to support reservoir operations using machine learning.
- Develop and implement modeling workflows incorporating ensemble forecasts into robust Forecast-Informed Reservoir Operations (FIRO).
- Collaborate with interdisciplinary teams to connect hydrologic modeling with operational water infrastructure.
- Contribute to peer-reviewed publications, technical reports, and present research at scientific and stakeholder-focused events.

Postdoctoral Research Scholar — **Aug 7, 2023 – January 1, 2025**
Arizona State University — Tempe, AZ, U.S.
Center for Hydrologic Innovations | School of Sustainable Engineering & Built Environment (SSEBE)

- Led the state-wide evaluation of NOAA's National Water Model (NWM) performance using in-situ observations across multiple scales in Arizona to support water resources management.
- Investigated the performance and sources of model uncertainty in the context of Arizona's hydroclimatology.
- Contributed to collaborative research outputs, peer-reviewed publications, conference presentations, and student supervision.

Technical Research Associate — **Apr 1, 2023 – Jul 31, 2023**
The University of Tokyo — Tokyo, Japan
Hydrodynamics Lab (Yamazaki Lab) | Institute of Industrial Science (IIS)

- Contributed to the development and implementation of a data assimilation scheme for the CaMa-Flood global hydrodynamic model, including authoring the official user manual.
- Assisted in designing and refining a benchmarking toolkit for evaluating CaMa-Flood model performance.

Project Researcher — **Apr 1, 2021 - Mar 31, 2023**
The University of Tokyo — Tokyo, Japan
Center for Global Commons (CGC) | Institute for Future Initiatives (IFI)

- Led research assessing the value of hydrometeorological forecasts for hydropower reservoir operations using a physics-based distributed hydrological model.
- Improved the representation of snow processes in the land surface model (LSM) scheme within the Water and Energy Budget-based Distributed Hydrological Model (WEB-DHM) using reanalysis datasets.
- Managed laboratory activities, partially supervised graduate students, taught courses, authored academic articles, and presented research at scientific conferences.

Project Researcher — **Oct 1, 2020 - Mar 31, 2021**
The University of Tokyo — Tokyo, Japan
River & Environmental Engineering Laboratory (REEL) | Department of Civil Engineering

- Research on developing an auto-correction scheme for Japan Meteorological Agency's (JMA) gridded radar precipitation product using WEB-DHM and MODIS snow observations.
- Development of a hydrometeorological forecasting system (comprehensive data pipeline development from scratch) in Japan using gridded weather forecasts, radar observations, satellite observations and a distributed hydrological model using the state-of-the-art Data Integration and Analysis System (DIAS) at The University of Tokyo.
- Management of laboratory activities, partial supervision of graduate level students, writing academic articles and presenting in scientific conferences.

Visiting Foreign Researcher — **Feb 19, 2018 - Mar 30, 2018**
Public Works Research Institute — Tsukuba, Japan
International Center for Water Hazard and Risk Management under the auspices of UNESCO (ICHARM)

- Learned the model physics, algorithms and numerical schemes employed in the Water and Energy Budget-based Distributed Hydrological Model (WEB-DHM) for future contribution to model development.

Education

The University of Tokyo | Department of Civil Engineering — **Sep 2017 – Sep 2020**
Ph.D. in Civil Engineering — Tokyo, Japan

The University of Tokyo | Department of Civil Engineering — **Oct 2015 – Sep 2017**
Masters in Civil Engineering — Tokyo, Japan

University of Engineering & Technology, Taxila | Department of Civil Engineering — **Nov 2011 – Jul 2015**
Bachelors of Science (Hons) in Civil Engineering | CGPA: 3.99/4.00 | Gold Medalist — Taxila, Pakistan

Technical Skills

- **Hydrological modeling:** WEB-DHM (development), VIC (application), NOAA National Water Model/WRF-Hydro (benchmarking), CaMa-Flood (application, benchmarking), model calibration and optimization, real-time flood forecasting.
- **Computing skills:** Python (xarray, pandas, geopandas and other geospatial packages), Fortran (OpenMP), UNIX/Linux shell scripting, parallel programming. Using HPC clusters
- **Geographic Information Systems:** ArcGIS, ArcGIS Pro (ArcPy), QGIS.
- **Cloud platforms:** Amazon Web Services (AWS), Cyverse, Google Cloud Platform (GCP).

Achievements

Awards

Best Lightning Talk Award | ASU Flow 2023, Arizona, U.S. — **2023**
- Best lightning talk award presented by ASUs Center for Hydrologic Innovations and Arizona Hydrological Society for talk titled 'Can We Improve Process-based Snow Hydrological Modeling by Using Reanalysis Vertical Temperature Profiles?'.

Excellent Young Researcher Award | World Bosai Forum, Sendai, Japan — **2017**
- Excellent Young Researcher Award for dedication to public safety as a speaker at the 16th International Symposium on New Technologies for Urban Safety of Mega Cities in Asia as a part of World Bosai Forum.

Nishino Akiyo Award | The University of Tokyo, Tokyo, Japan — **2016**
- Recognized for top performance in the year-long Japanese Language Course.

Four Gold Medals for B.S. in Civil Engineering | University of Engineering & Technology, Taxila, Taxila, Pakistan — **2015**
- Awarded four gold medals: (i) UET Taxila Gold Medal, (ii) National Development Consultant Gold Medal, (iii) Savola Gold Medal, & (iv) Crescent Gold Medal for overall best performance in B.S. Civil Engineering.

Best Dissertation Award for B.S. in Civil Engineering | University of Engineering & Technology, Taxila, Taxila, Pakistan — **2015**
- Received first place and a 15,000 PKR award for undergraduate research dissertation on GIS-based hydropower potential evaluation at Open House 2015.

Best Performance Award | Water And Power Development Authority, Faisalabad, Pakistan — **2014**
- Achieved first position in the Summer Internship Program out of 31 participants.

National Recognition | National Engineering Services Pakistan Pvt. Limited, Islamabad, Pakistan — **2011**
- Featured in NESPAK's September 2011 newsletter for achieving the highest rank in the Engineering College Admission Test (ECAT) among 31,553 candidates nationwide.

Scholarships

Monbukagakusho (MEXT) | The University of Tokyo, Tokyo, Japan — **2015 – 2020**
- The Ministry of Education, Culture, Sports, Science and Technology (MEXT) of Japan, Scholarship for Masters and PhD Degree (fully-funded) at The University of Tokyo, Japan.

Best Performance Scholarship | University of Engineering & Technology, Taxila, Pakistan — **2011 – 2015**
- Grant amounting to 70,000 PKR awarded for securing 1st position through 7 semesters of B.S. Civil Engineering.

Federal Board of Intermediate & Secondary Education | University of Engineering & Technology, Taxila, Pakistan — **2011 – 2015**
- Received merit-based scholarship of 12,000 PKR annually for academic excellence during undergraduate studies.

Publications

Peer-reviewed articles

1. **Moiz, A.**, Kawasaki, A. (2024), Long-range streamflow prediction using a distributed hydrological model in a snowfed watershed, *Proceedings of IAHS*, *386*, 217–222. DOI
2. Ashraf, S., Ali, M., Shrestha, S., Hafeez, M. A., **Moiz, A.**, Sheikh, Z. A. (2022), Impacts of climate and land-use change on groundwater recharge in the semi-arid lower Ravi River basin, Pakistan, *Groundwater for Sustainable Development*, *17*, 100743. DOI
3. **Moiz, A.**, Wei, Z., Naseer, A., Kawasaki, A., Acierto, R. A., Koike, T. (2022), Improving Snow-Process Modeling by Evaluating Reanalysis Vertical Temperature Profiles Using a Distributed Hydrological Model, *Journal of Geophysical Research: Atmospheres*, *127*(18), e2021JD036174. DOI
4. Hafeez, M. A., Nakamura, Y., Suzuki, T., Inoue, T., Matsuzaki, Y., Wang, K., **Moiz, A.** (2021), Integration of Weather Research and Forecasting (WRF) model with regional coastal ecosystem model to simulate the hypoxic conditions, *Science of the Total Environment*, *771*, 145290. DOI
5. Tamakawa, K., Mohamed, R., Naseer, A., Ushiyama, T., Nakamura, S., Nyunt, C. T., **Moiz, A.**, Onuma, K., Koike, T. (2021), Investigation of ensemble dam inflow simulation in Sai River basin, *Journal of Japan Society of Civil Engineers, Ser. B1 (Hydraulic Engineering)*, *77*(2), I_61–I_66. DOI
6. Thin, K. K., Zin, W. W., San, Z. M. L. T., Kawasaki, A., **Moiz, A.**, Bhagabati, S. S. (2020), Estimation of run-of-river hydropower potential in the Myitnge river basin, *Journal of Disaster Research*, *15*(3), 267–276. DOI
7. Wei, Z., He, X., Zhang, Y., Pan, M., Sheffield, J., Peng, L., Yamazaki, D., **Moiz, A.**, Liu, Y., Ikeuchi, K. (2020), Identification of uncertainty sources in quasi-global discharge and inundation simulations using satellite-based precipitation products, *Journal of Hydrology*, *589*, 125180. DOI
8. **Moiz, A.**, Kawasaki, A., Koike, T., Shrestha, M. (2018), A systematic decision support tool for robust hydropower site selection in poorly gauged basins, *Applied Energy*, *224*, 309–321. DOI

Manuscripts (in-prep or under review)

1. **Moiz, A.**, Kawasaki, A. (under-review), Value of a real-time ensemble short-range hydrometoerological prediction system for improved performance of a hydropower dam cascade.
2. **Moiz, A.**, Mascaro, G. (under-review), A multiscale evaluation of the water balance components and their uncertainty in Arizona using the National Water Model.
3. Zhang, Z., Kawasaki, A., **Moiz, A.** (under-review), Development of a method for assessing the impact of climate change on hydropower dams based on a parameterized approach.

Datasets

1. **Moiz, A.** (2025). Arizona Hydrologic Fluxes (1981–2020) – National Water Model, HydroShare, http://www.hydroshare.org/resource/197f5389e8844a6f9cce31c239f81f08

Presentations

Conferences

1. Raney, K. S., Erfani, M., **Moiz, A.** (Apr. 2026), Tradeoffs between forecast ensemble dimension and multi-objective reservoir performance using scenario reduction and tree-based model predictive control, *World Environmental & Water Resources Congress*.
2. **Moiz, A.**, Raney, K. S., Kalansky, J., Brandt, T., Pan, M., Hartman, R., Dixon, T. (Dec. 2025), Adaptive reservoir operations in a snow-dominated basin using direct policy search, *American Geophysical Union Fall Meeting*.
3. Parajuli, B., Mascaro, G., **Moiz, A.** (Dec. 2025), Evaluation and improvment of precipitation products in Arizona supporting water resources planning and management, *American Geophysical Union Fall Meeting*.
4. Zhang, Z., Kawasaki, A., **Moiz, A.** (Dec. 2025), Physics-informed and interpretable LSTM for hydropower reservoir release modeling under climate variability, *American Geophysical Union Fall Meeting*.
5. **Moiz, A.**, Gupta, A. (Mar. 2024), Meixner tri-university legacy project: approach to data access, hydroclimate modeling and scenario development, progress to date, *Water Resources Research Center 2024 Annual Conference*. URL
6. **Moiz, A.**, Mascaro, G. (Mar. 2024), A reconstruction of the water balance at key basins in Arizona using a high-resolution hydrologic reanalysis, *Water Resources Research Center 2024 Annual Conference*. URL
7. Qiu, Y., Niu, G.-Y., Behrangi, A., Famiglietti, J. S., Farmai, M. A., Gupta, A., Yousefi, H., Jawad, M., **Moiz, A.**, Mascaro, G., Gupta, N., Jacobs, K. (Dec. 2024), The significant impact of precipitation intensity on natural groundwater recharge and terrestrial water storage change in arid regions, *American Geophysical Union Fall Meeting*. URL
8. Yamazaki, D., Revel, M., **Moiz, A.**, Ikeshima, D., Kanae, S. (July 2024), Assimilation of satellite observed river hydrodynamics to a global river model, *9th Global Energy and Water EXchanges (GEWEX) Open Science Conference*. URL

9. Zhang, Z., Kawasaki, A., **Moiz, A.** (Dec. 2024), Assessing the impact of climate change on hydropower: Operational strategies for hydropower dams using a parameterized approach, *American Geophysical Union Fall Meeting*. **URL**
10. **Moiz, A.** (Nov. 2023), Can we improve process-based snow hydrological modeling by using reanalysis vertical temperature profiles, *ASU Flow 2023*. **URL**
11. **Moiz, A.**, Kawasaki, A. (Feb. 2023), Long-range streamflow prediction using a distributed hydrological model in a snowfed watershed, *The 9th International Conference on Flood Management*. **URL**
12. **Moiz, A.**, Kawasaki, A. (Dec. 2021), Evaluating the skill of a seasonal scale hydrometeorological prediction system in Japan based on downscaled meteorological forecasts, *American Geophysical Union Fall Meeting*. **URL**
13. Hafeez, M. A., Nakamura, Y., Inoue, T., Matsuzaki, Y., **Moiz, A.** (Dec. 2019), Modeling coastal ecosystem of a semi enclosed water body by coupling the regional weather model WRF to ecosystem model, *American Geophysical Union Fall Meeting*. **URL**
14. **Moiz, A.**, Wei, Z., Acierto, R. A., Naseer, A., Kawasaki, A., Koike, T. (Dec. 2019), Modeling seasonal snow hydrology using vertical air temperature profiles from reanalysis, *American Geophysical Union Fall Meeting*. **URL**
15. **Moiz, A.**, Kawasaki, A. (May 2018), Representing hydropower dam operations in a physically-based distributed hydrological model, *Japan Geoscience Union Meeting*. **URL**
16. **Moiz, A.**, Kawasaki, A. (Dec. 2018), Understanding the reliance of Kurobe River's cascaded dam system on seasonal snowmelt runoff using a distributed hydrological model, *American Geophysical Union Fall Meeting*. **URL**
17. **Moiz, A.**, Kawasaki, A. (July 2017), A GIS based decision support tool for hydropower planning, *Esri User Conference*. **URL**
18. **Moiz, A.**, Kawasaki, A. (Nov. 2017), Cascaded hydropower scheme operation in a geospatial site selection framework, *16th International Symposium on New Technologies for Urban Safety of Mega Cities in Asia*. **URL**
19. **Moiz, A.**, Kawasaki, A. (Mar. 2017), Development of an integrated methodology for run-of-river hydropower resources assessment using GIS and advanced distributed hydrological modelling, *Monitoring of Global Environment and Disaster Risk Assessment from Space : the IIS Forum proceedings*. **URL**
20. **Moiz, A.**, Kawasaki, A., Koike, T. (Oct. 2016), A GIS based approach for hydropower site selection and potential evaluation of Kunhar River Basin, Pakistan, *International Association of Geo-informatics*. **URL**

Invited talks

1. **Moiz, A.** (Dec. 2019), Integrating geospatial technologies with advanced hydrological models for robust energy planning, *Center for Geographic Analysis, Harvard University*. **URL**

REFERENCES

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